Employee–Project Allocation System

Java OOP + 2D Array Descriptive Problem

Problem Statement

A software company manages several employees who are working on different projects. The company wants to maintain the number of hours each employee has worked on each project. The data is stored in a 2D array where each row represents an employee and each column represents a project.

The company now wants to generate a project-wise report from the employee-wise data. To do this, the program must transform the original matrix into its transpose, so that rows represent projects and columns represent employees.

Requirements

1. Create a class **ProjectAllocation**.

2. Store employee-project working hours using a 2D integer array named **hours**.

3. Create a constructor to initialize the employee-project data.

4. Create **displayEmployeeReport()** to display the original employee-wise working hours.

5. Create **calculateEmployeeTotal()** to calculate the total hours worked by each employee.

6. Create **generateProjectReport()** to generate and display the transpose of the matrix.

7. Create **calculateProjectTotal()** to calculate the total hours spent on each project.

8. Create **findMostWorkedProject()** to identify the project with the maximum total working hours.

Sample Data

	P1	P2	P3
Employee E1	20	15	10
Employee E2	12	18	14
Employee E3	25	10	20
Employee E4	15	22	12

Expected Project-wise Report (Transpose)

	E1	E2	E3	E4
Project P1	20	12	25	15
Project P2	15	18	10	22
Project P3	10	14	20	12

Expected Calculations

• Employee E1 Total: 45 hours

• Employee E2 Total: 44 hours

• Employee E3 Total: 55 hours

• Employee E4 Total: 49 hours

• Project P1 Total: 72 hours

• Project P2 Total: 65 hours

• Project P3 Total: 56 hours

• Most Worked Project: P1

OOP and Programming Conditions

1. Use class and objects.

2. Use a constructor.

3. Use encapsulation through appropriate fields and methods.

4. Use methods for each major operation.

5. Use a 2D array for storing the data.

6. Perform matrix transpose manually using nested loops.

7. Do not use ArrayList, Collections, or any built-in matrix operation.

Student Task

Write the complete Java program based on the above requirements. The program should create an object of ProjectAllocation, initialize the matrix, display the employee report, calculate employee totals, generate the project-wise transpose report, calculate project totals, and identify the most worked project.